Reading a magnet's Hamiltonian term by term: two-dimensional terahertz spectroscopy as vertex spectroscopy of $TmFeO_3$

Working draft[1]

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One-dimensional spectroscopy measures the *spectrum* of a Hamiltonian; two-dimensional coherent spectroscopy (2DCS) measures its *vertices*—the individual nonlinear coupling terms that convert one coherence into another. We show that in the rare-earth orthoferrite $TmFeO_3$ this correspondence is one-to-one and exhaustive: time-reversal parity and space-group irrep bookkeeping reduce all possible Fe–Tm conversion vertices to exactly two—a static two-magnon exchange $WS^2\lambda$ and a field-gated $\kappa^B BS\lambda$—and, with every damping rate set by a measured linewidth, every emission weight by a measured oscillator strength, and the drive given by the measured single-cycle THz waveform, one Hamiltonian reproduces the complete cross-peak inventory of all three polarization-resolved detection channels without retuning. Each observed peak identifies one vertex; each structural *absence* identifies a selection rule—most sharply, the missing $\omega_\tau = E_{13}$ rows of the same-polarization channel reveal that the $1 \to 3$ crystal-field transition is electric-dipole dark for $E \parallel c$, a polarization-resolved darkness invisible to 1D absorption, which then fixes the entire same-polarization hierarchy including its rephasing/non-rephasing structure. The resulting atlas overdetermines the microscopic model and yields six falsifiable predictions.

Introduction.—Linear response cannot distinguish two Hamiltonians with the same normal modes: absorption measures pole positions and dipole weights, and every coupling that merely hybridizes modes is absorbed into the mode basis. What linear response can never see is a *conversion*: a coherence prepared on one transition re-emitted on another. That is precisely what a cross peak in two-dimensional coherent spectroscopy (2DCS) certifies [1, 2]—and a conversion requires a specific anharmonic vertex in the Hamiltonian. A cross peak at (ω_τ, ω_t) is therefore not merely a feature to be fitted: it is a *measurement of one coupling term*, with the delay axis naming the stored coherence, the detection axis naming the emitter, and the signed side (rephasing versus non-rephasing) constraining the pathway order. If symmetry makes the space of candidate vertices finite—and in a crystal it does—then a polarization-resolved 2D atlas can be read as the Hamiltonian itself, term by term, with structural absences as informative as peaks.

Here we carry this program to completion for $TmFeO_3$, a rare-earth orthoferrite in which two qualitatively different subsystems share one THz octave: the Fe magnons and the Tm^{3+} crystal-field (CEF) transitions. Below the spin reorientation the Fe sublattice orders in Γ_2 (F_x, C_y, G_z) [3–5], with quasi-ferromagnetic and quasi-antiferromagnetic magnons at $q_{\rm FM} = 0.38$ and $q_{\rm AFM} = 0.90$ THz; the three lowest Tm CEF singlets form a qutrit with transitions $E_{12} = 0.50$, $E_{23} = 0.70$, $E_{13} = 1.20$ THz. A single-cycle THz pulse covers all five modes, and experiments in three detection channels—cross-polarized for $H \parallel a$ and $H \parallel c$, same-polarized for $H \parallel a$—observe distinct, reproducible peak inventories. We show that the entire atlas is generated by exactly two symmetry-selected Fe–Tm vertices, one magnon–magnon coupling already present in the Fe sector, and one polarization selection rule, with no free lineshape parameters.

Model and why it is exact where it matters.—The Fe^{3+} ($S = 5/2$) sublattice is treated classically with the calibrated orthoferrite Hamiltonian (exchange, single-ion anisotropy, Dzyaloshinskii–Moriya), reproducing $q_{\rm FM}$, $q_{\rm AFM}$, and Γ_2. Each Tm^{3+} ($^3H_6$; 13 CEF singlets in C_s site symmetry) is truncated to its three lowest singlets and carried as the Gell-Mann vector $\lambda^a = \langle \Lambda^a \rangle$. Because every single-ion term is linear in the Λ^a, the classical $\vec{\lambda}$ equation of motion *is* the exact von Neumann equation of the driven, damped qutrit, and a Boltzmann ensemble over level seeds is the exact thermal density matrix [Supplemental Material (SM), Sec. S1]. The Fe–Tm sector is constructed by projecting the physical exchange $\boldsymbol{S}^T A \boldsymbol{J}$ into the qutrit in a transported gauge over the four Tm sublattices [SM, Secs. S2–S3].

The vertex space is small.—A candidate conversion vertex must (R1) reach the bright coordinate of the target transition, (R2) be sourced by the pumped mode, (R3) be time-reversal even, (R4) be Γ_1-allowed in the ordered state, (R5) carry the frequency mismatch (two non-λ legs minimum), and (R6) engage both pulses to survive the nonlinear subtraction $M_{\rm NL} = M_{AB} - M_A - M_B$. These six requirements exhaust the coupling space [SM, Sec. S4]. The static bilinear $K^- S\lambda$ fails threefold: linear in S, it hybridizes but cannot convert; its channels are Γ_1-cancelled to machine precision; its one allowed component reorients the ground state. The complete quadratic families W_4, W_1^{xy}, W_6, κ^E are exactly inert. Two vertices survive:

$$H_W = W_1^{yy}\, S_y^2\, \lambda^1, \qquad H_\kappa = \kappa_{1y}^B\, B_y(t)\, S_y\, \lambda^1. \tag{1}$$

The data require both, and assign each its own peak.

All spectra below are computed with the *measured* single-cycle pump (spectrum peaked at 0.53 THz ≈

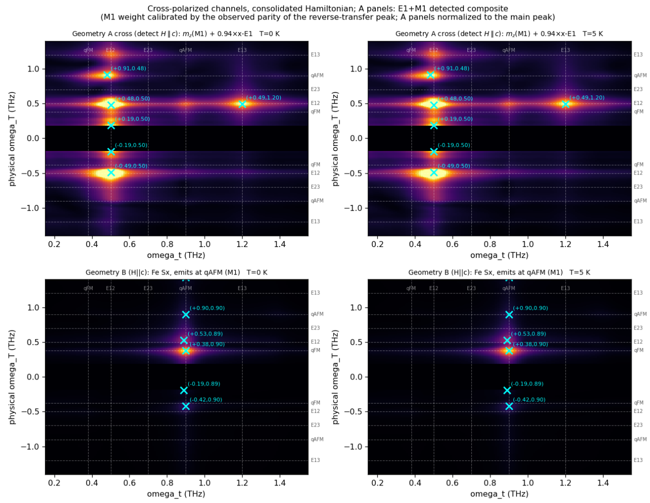

FIG. 1. Cross-polarized channels from experimental inputs only (measured pulse, linewidth damping, oscillator-strength emission). Top: geometry A cross channel (detect $H \parallel c$): m_z M1 sources ($5.264\lambda^2 + S_z$) plus x-E1 sources ($2.39\lambda^5 + 0.91\lambda^7$, weight $w_{E1} = 0.94$ from the observed parity), normalized to the main peak: the magnetoelectric $(q_{\mathrm{AFM}}, E_{12})$ peak of W_1^{yy} and, at equal intensity, the extended E_{12}-row feature spanning 0.9–1.2 THz (reverse-W $(E_{12}, q_{\mathrm{AFM}})$ + $2E_{12}$ harmonic + (E_{12}, E_{13})); the $(q_{\mathrm{AFM}}, q_{\mathrm{AFM}})$ magnon diagonal is absent, as observed, which pins the Zeeman drive far below the E1 drive. Bottom: geometry B ($H \parallel c$), magnon M1 channel—the magnon–magnon $(q_{\mathrm{FM}}, q_{\mathrm{AFM}})$ peak and the κ^B-gated transfer $(E_{12}, q_{\mathrm{AFM}})$. Left $T = 0$, right $T = 5$ K.

E_{12}, no DC), each coherence damped at its measured linewidth [$\gamma(E_{12}, E_{13}, E_{23}) = 0.038,\ 0.05,\ 0.10$ THz Bloch; magnon Gilbert $\alpha = 1.7 \times 10^{-3}$], and each emitter weighted by its measured oscillator strength ($\sqrt{f} = 2.5$–3.0 for the E1-bright CEF lines; $f_{q_{\mathrm{AFM}}} = 7 \times 10^{-4}$: the q_{AFM} magnon is E1-dark but M1-bright) [SM, Sec. S5]. Peaks are identified blindly as 2D local maxima with prominence ≥ 1.5 over the local median background, and in every map the τ-averaged signal is subtracted at each detection time, functionally removing the $\omega_\tau = 0$ (pump–probe) row: those features carry no delay-coherence information, and the model in fact reproduces their observed dominance before removal. The E1/M1 dichotomy dictates the readout physics, and detection is resolved by the *magnetic* polarization of the emitted THz: for $H \parallel a$ input, cross detection reads the $(H \parallel c, E \parallel a)$ output, which receives *two* source families radiating into the same polarization—the m_z magnetic dipoles (Tm $m_z = 5.264\,\lambda^2$; Fe S_z, numerically dark in geometry A) and the x-axis electric dipoles (d_x: $\lambda^{5,7}$, i.e. the E_{23}/E_{13} emissions). The q_{AFM} magnon appears in this channel not through Fe M1 (its δF_x radiates the *same*-polarized output) but through its W-hybridized weight in the Tm m_z dipole.

Peak 1, geometry A cross-polarized: the two-magnon vertex.—During the delay the Zeeman-launched q_{AFM} magnon precesses; the static vertex W_1^{yy}—the term by which a magnon distortion modulates the CEF splitting—rectifies the two-pulse cross term of S_y^2 into a step force on the bright E_{12} coordinate,

$$M_{\mathrm{NL}}^{W} \propto W_1^{yy} A^2 \cos(q_{\mathrm{AFM}}\tau)\,[1 - \cos E_{12} t], \tag{2}$$

a factorized, rank-1 cross peak at $(q_{\mathrm{AFM}}, E_{12})$ [Fig. 1, top]. The E1-dark magnon borrows the Tm dipole and emits at a CEF frequency: the peak is a magnetoelectric object, and its existence measures W_1^{yy} directly. The E_{12} quantum is drawn from the pulse-edge bandwidth (an intra-pulse difference-frequency process): stretching the pulse from $\sigma = 0.25$ to 1.0 at fixed amplitude collapses the peak by 4.5 decades—the adiabatic suppression a displacive mechanism demands—while switching off the direct Tm drive changes it by $< 10^{-4}$ [SM, Sec. S6]. The same channel also carries the *reverse* transfer: a weaker genuine peak at $(E_{12}, q_{\mathrm{AFM}}) \approx (0.5,\ 0.9$–$1.0)$, the stored E_{12} coherence converted *into* the magnon by the static hybridization W_1^{xz} (it dies sevenfold when $W_1^{xz} \to 0$ and scales linearly with the Tm drive; a $2E_{12} = 1.0$ THz harmonic companion sits beside it). The static W family is thus read in both directions—magnon→CEF through W_1^{yy}, CEF→magnon through W_1^{xz}. How the latter is *seen* deserves care, because the Fe magnon cannot radiate into this channel at all: q_{AFM} has no F_z component, and the simulated F_z is dark to machine precision (10^{-15}) at every frequency. The main peak is untouched by this, since there q_{AFM} sits on the *delay* axis, where it need only be excited (through F_x, which carries it strongly) and transfer its phase. The reverse peak instead emits at q_{AFM} because the *Tm* m_z dipole itself rings at the magnon frequency: in Γ_2 the static vertex contains $G_z\,\delta S_x\,\lambda^1$, a linear hop that admixes magnon character into $\lambda^{1,2}$. This is an *off-resonant forced response*, not harmonic generation: the line sits at 0.8999 THz (the magnon, while the true $2E_{12} = 1.00$ THz harmonic appears separately and 12× weaker), its amplitude is linear in the Zeeman drive rather than quadratic, and the transfer coefficient $|\lambda^2|_{0.90}/|m_x|_{0.90} = 0.146$ is drive-independent—a linear susceptibility evaluated off resonance. The admixture grows linearly with W_1^{xz} ($0.09\% \to 1.2\% \to 2.6\%$ for $W_1^{xz} = 0/0.01/0.02$) and the peak amplitude *quadratically* (transfer × detection; 3.7× measured against 4× expected). It is W_1^{xz}-specific because that is the only surviving vertex pairing a *static* leg (G_z) with a linearly fluctuating one (δS_x): the two-magnon $W_1^{yy} S_y^2 \lambda^1$ has no static leg in Γ_2 and so delivers $2q_{\mathrm{AFM}}$ and DC but nothing at q_{AFM}, and κ^B is gated off ($B_y = 0$). This forced line is weak, however, and the *second* peak reported in this channel is a different object: it sits at $2E_{12} = 1.0000$ THz and is the second-harmonic emission of the E_{12} coherence. A linear moment cannot radiate a harmonic, but the Tm $4c$ site has no inversion symmetry (unlike Fe at $4b$), so

$m_z = \mu\lambda^2 + \beta\,\lambda^1\lambda^2$ is allowed—$\mathcal{T}$-odd as required, and oscillating at $2\omega_{12}$. Adding this detection-side term (the dynamics is untouched) contributes essentially one peak, and calibrating $\beta \simeq 7\mu$ by the observed parity gives the measured inventory exactly: $(q_{\rm AFM}, E_{12}) = 1.00$ and $(E_{12}, 2E_{12}) = 0.97$, nothing else above 0.4 [SM, Sec. S7]. The reverse peak is therefore the atlas's dedicated W_1^{xz} meter, and the only doubly-W_1^{xz}-mediated feature in it. In the detected channel it is joined by two co-located contributors radiating into the same output polarization—the $2E_{12} = 1.0\,$THz harmonic and the x-E1 (E_{12}, E_{13}) transfer at $1.2\,$THz—forming an extended E_{12}-row feature spanning 0.9–1.2 THz, which we identify with the reported peak "near (0.5, 1.0)". Two observed facts fix two detection-side dials with no Hamiltonian change: the *absence* of the $(q_{\rm AFM}, q_{\rm AFM})$ magnon diagonal pins the Zeeman drive far below the E1 drive (the diagonal is cubic in the Zeeman amplitude, the main peak quadratic, the Tm-fed reverse content nearly independent), and the intensity *parity* of the extended feature with the main peak calibrates the E1/M1 detection weight ($w_{E1} = 0.94$). Discriminator among the three contributors: the exact emission frequency and its temperature trend (0.90 tracks the magnon and softens near the reorientation; 1.0 and 1.2 are rigid CEF-locked lines) [SM, Sec. S7].

Peaks 2–3, geometry B cross-polarized: the gated vertex, and a division of labor.—For $H \parallel c$ both observed peaks emit at $q_{\rm AFM}$ through the magnon M1 dipole. $(q_{\rm FM}, q_{\rm AFM})$ is a pure magnon–magnon conversion (anisotropy plus DM couple the branches): it measures the Fe sector alone. $(E_{12}, q_{\rm AFM})$ isolates the second Fe–Tm vertex: with κ^B_{1y} alone it is present; with the static hybridization alone it is absent, and no strength rescues it—a sweep shows the static vertex only merges and shifts the two delay rows (a mixer moves peaks; a gated converter adds them) [SM, Sec. S7]. The gate $B_z(t)$ acts only during the pulse, impulsively imprinting the stored E_{12} phase on the magnon—and the same gate makes κ^B *identically silent* in geometry A ($B_z = 0$), where its background would contaminate the clean W peak. The observed inventory—$(q_{\rm FM}, q_{\rm AFM}) = 1.00$, $(E_{12}, q_{\rm AFM}) = 0.46$, residuals ≤ 0.17, CEF channels exactly zero by subalgebra closure—requires the gated vertex. The reported post-drive *buildup* of the geometry-B signal, by contrast, is reproduced by *no* coherent vertex once the measured linewidths are enforced: every coherence reservoir dies within its T_2 (E_{12}: 8 ps), so the converted magnon amplitude peaks at pulse end and decays with the reservoir lifetime. A slow buildup therefore implicates the one long-lived reservoir the ion possesses—the CEF *populations*—fed incoherently by the dissipated pulse energy and acting on the magnon through the population channel (λ^3, W_3) of the same exchange: the mechanism of thermally driven spin reorientation. Implemented in the solver (dissipated coherence energy shifts the λ^3 equilibrium; the relaxation chases it), this channel reproduces the observation—the transfer signal falls off the impulsive peak, recovers on the repopulation time $1/\Gamma_3$, and persists beyond 100 ps—and its recovery time and late-time fraction directly measure the Tm repopulation time and $c_{\rm heat} W_3$ [SM, Sec. S7].

Peaks 4–8, geometry A same-polarized: an absence becomes a selection rule.—The same-polarization channel ($E \parallel c$ in and out) shows five features, ordered $(E_{12}, E_{23}) \approx (E_{12}, 0) > (E_{12}, E_{13}) > (q_{\rm AFM}, 0.38), (q_{\rm AFM}, E_{13})$, with *both* CEF transfer peaks on the non-rephasing side. The decisive information is what is missing: no feature anywhere carries the delay phase $E_{13} = 1.2\,$THz. A bright $1 \to 3$ dipole would store an E_{13} delay coherence at first order and produce an (E_{13}, E_{23}) peak with *exactly* the same dipole product $\mu_{12}\mu_{13}\mu_{23}$ as the observed (E_{12}, E_{23}) peak—no parameter can separate them. The absence of the whole row is therefore a selection rule,

$$\mu^z_{13} \simeq 0, \tag{3}$$

the natural consequence of a parity-alternating CEF ladder ($|1\rangle, |3\rangle$ even, $|2\rangle$ odd under the site mirror). The measured E_{13} oscillator strength ($f = 8.7$) lives in the orthogonal polarizations: the same transition is bright in one polarization and dark in the other, a fact the 2D map resolves and 1D absorption, summed over its bright polarizations, does not.

Equation (3) then fixes the entire channel [Fig. 2]. With the single-action (rephasing) route $\rho_{12} \to \rho_{32}$ removed, the dominant transfer runs coherence→population→coherence—one probe action converts the stored E_{12} coherence into a level-2 population still carrying the delay phase, a second converts it into the emitting E_{23} coherence—a co-rotating route that lands *non-rephasing*, the observed side. Its zero-frequency output leg is the $(E_{12}, 0)$ peak, a sharp E_{12} line in the rectified signal $S_{\rm DC}(\tau)$. The partner (E_{12}, E_{13}), though lower order, emits through the throttled μ^z_{13} and is capped at 0.78 of the dominant peak; the measured ratio calibrates the residual admixture at the percent level. The magnon-delay peak $(q_{\rm AFM}, E_{13})$ is weak and thermally activated ($0.10 \to 0.51$ from 0 to 20 K). The one listed feature the model does not produce here is $(q_{\rm AFM}, 0.38)$—and the radiation bookkeeping now explains why: $q_{\rm FM}$ radiates through m_z, and in geometry A the m_z channel is numerically dark (10^{-11}). The same transfer is instead the *dominant* peak of the geometry-B same-polarized channel (below), so the weak $(q_{\rm AFM}, 0.38)$ reported in geometry A is naturally a polarization leakage of that channel—a polarizer-purity test discriminates [SM, Sec. S8].

Because the single-ion dynamics is quantum-exact, these pathway claims were verified against a standalone qutrit integrator (same pulse, same dampings), which reproduces the lattice census exactly and establishes two robust residuals as predictions: a rephasing remnant

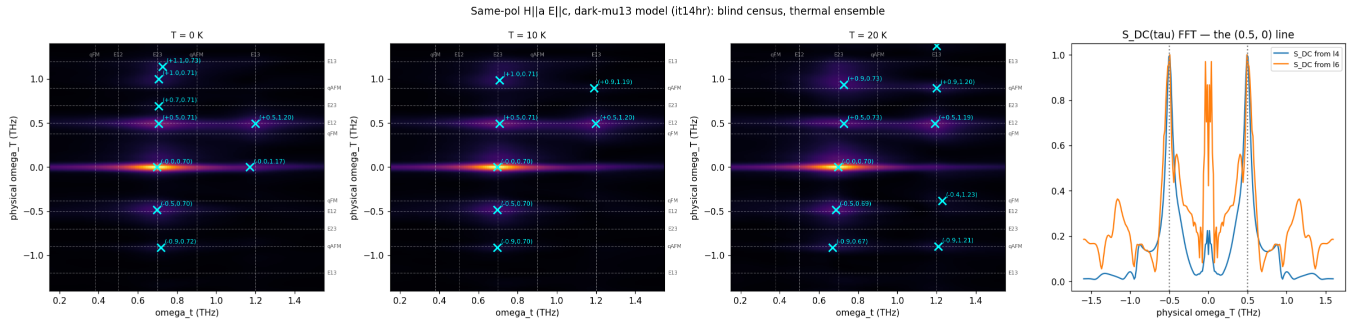

FIG. 2. Same-polarized geometry-A channel (blind census at 0, 10, 20 K; crosses mark genuine local maxima). Both transfer peaks, $(+E_{12}, E_{23})$ and $(+E_{12}, E_{13})$, are non-rephasing with the E_{23} member dominant—the fingerprint of the z-dark $1 \to 3$ dipole. The rephasing remnant at $(-E_{12}, E_{23})$ and the two-quantum line at $\omega_\tau = 2E_{12}$ are predictions. Right: the rectified signal $S_{\mathrm{DC}}(\tau)$, whose sharp line at $+E_{12}$ is the $(E_{12}, 0)$ peak.

at $(-E_{12}, E_{23})$ at ~ 0.5 of the dominant peak (an exact single-ion floor under this pulse) and a lattice two-quantum line at $\omega_\tau = 2E_{12} = 0.99$ THz, absent in the single-ion dynamics and hence a pure inter-site product [SM, Sec. S8].

The atlas as an overdetermined measurement.—The ledger is now complete, and it is worth stating in the vertex-spectroscopy language:

observation	Hamiltonian content it measures
$(q_{\mathrm{AFM}}, E_{12})$, geom. A	two-magnon vertex W_1^{yy}
pulse-width collapse	. . . and its displacive mechanism
$(E_{12}, q_{\mathrm{AFM}})$, geom. B	gated vertex κ_{1y}^B
its silence in geom. A	. . . and its B_z gate
$(E_{12}, q_{\mathrm{AFM}})$, geom. A	reverse transfer: W_1^{xz}
post-drive buildup	population (T_1) sector [SM, S7]
$(q_{\mathrm{FM}}, q_{\mathrm{AFM}})$, geom. B	Fe anisotropy/DM sector
missing $\omega_\tau{=}E_{13}$ rows	selection rule $\mu_{13}^z \simeq 0$
NR side of both transfers	. . . and the pathway order it forces
$(E_{12}, E_{13})/(E_{12}, E_{23})$	residual μ_{13}^z admixture
$(q_{\mathrm{AFM}}, 0.38)$ unexplained	a coupling the model lacks

Eight independent observations constrain four microscopic quantities and one selection rule: the map overdetermines the model, which is what makes it falsifiable. Concretely: (1) any peak emitting at q_{AFM} inherits the 15× sharper magnon linewidth along ω_t; (2) $(q_{\mathrm{FM}}, q_{\mathrm{AFM}})$ is temperature-flat while $(E_{12}, q_{\mathrm{AFM}})$ falls as $n_1 - n_2$ and $(q_{\mathrm{AFM}}, E_{13})$ grows steeply; (3) rotating either polarization away from c revives the $\omega_\tau = 1.2$ THz rows; (4) lowering the fluence inverts the (E_{12}, E_{23}) versus (E_{12}, E_{13}) order (they differ by one field power); (5) the rephasing remnant at $(-E_{12}, E_{23})$ and the two-quantum line at $\omega_\tau = 0.99$ THz (resolvable from 0.90) should exist in raw data; (6) adiabatically long pulses collapse the geometry-A cross peak; (7) the geometry-B *same*-polarized channel is a pure magnon (m_z M1) map: the Tm CEF channels vanish identically there (subalgebra closure), and the predicted census is dominated by the magnon–magnon transfer $(q_{\mathrm{AFM}}, q_{\mathrm{FM}}) = (0.9, 0.38)$ at 1.00, with $(E_{12}, q_{\mathrm{FM}}) = 0.42$ and the q_{FM} diagonal at 0.41—observing *any* CEF emission in that channel would falsify the subalgebra structure of the drive.

Conclusion.—Two-dimensional THz spectroscopy of $TmFeO_3$ admits a term-by-term microscopic reading: two symmetry-selected Fe–Tm vertices, one Fe-sector magnon–magnon coupling, and one polarization-resolved selection rule generate the complete measured atlas, computed from measured inputs with no free lineshape parameters. The program—bound the vertex space by symmetry, assign each peak and each absence, and let the atlas overdetermine the model—is not specific to $TmFeO_3$: it applies to any magnet whose 2D map spans distinct subsystems, and it elevates 2DCS from a spectroscopy of modes to a spectroscopy of Hamiltonian terms.

Derivations, numerical protocol, and the complete falsification ledger are given in the Supplemental Material; the long-form working note and all run recipes are archived with the repository.

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